YOLO-World (FP32)

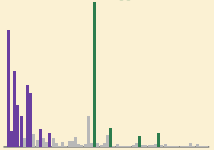


calibration images (256, COCO vocab.)

## D Leak-free diagnosis

quantize one conv to W4 (rest W8A8), measure COCO top-1 flip vs. FP

12.cv2: 31 pp



flip increase per conv (YOLO-World-S)

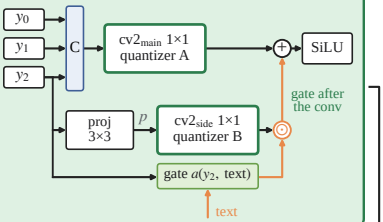
text-gated fusion convs → G
(all T-CSPLayer cv2, by structure)

other top convs → P
(greedy by damage, 1.5% budget)

## G Gate commutation

*FP-equivalent · +0 bits*

in every T-CSPLayer (12, 15, 19, 22)




## P Protection

YOLO Backbone (blocks 0–9)




activations stay low-bit (A4–A8)

* stem is W8 by protocol (first layer), not by P

**two candidates**

same seed · same calibration images

### GP

no M → calibrate + BRECQ (R)

BRECQ or QDrop

### GPM

M producer: $x/s$ consumers: $W\mathrm{diag}(s)$ → calibrate + BRECQ (R)

shared $s$ per producer

## M check



held-out train images (next 200) + COCO vocab.

top-1 flip vs. FP

$flip_{GPM} \le flip_{GP}$ → GPM, else GP

Ours

low-bit model

per seed